STUDENT PASS PREDICTION SYSTEM

(Project Report)

Submitted by:

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Course: Artificial Intelligence and Machine Learning

1. Introduction

In today's education system, predicting student performance can help educators take early action to improve outcomes. This project focuses on building a machine learning model that predicts whether a student will pass or fail based on academic and behavioral factors.

The system uses a Decision Tree Classifier, a supervised learning algorithm, to analyze student data and generate predictions.

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2. Objectives

- To develop a predictive model for student performance
- To analyze academic factors affecting results
- To visualize decision-making using a decision tree
- To evaluate model performance

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3. Problem Statement

Educational institutions often struggle to identify students at risk of failing. This system helps:

- Detect weak students early
- Improve teaching strategies
- Increase pass rates

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4. Dataset Description

A synthetic dataset of 200 students was created with:

- Hours Studied
- Attendance
- Assignments Completed
- Class Participation
- Pass (Target Variable)

Pass Criteria

- Total score > 22
- Attendance > 65%

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5. Methodology

Step 1: Data Generation

Data generated using NumPy and Pandas

Step 2: Data Preprocessing

- Feature selection
- Train-test split (80/20)

Step 3: Model Selection

- Decision Tree Classifier
- Max depth = 3

Step 4: Training

Model trained on training dataset

Step 5: Prediction

Predictions made on test dataset

Step 6: Evaluation

- Accuracy
- Confusion Matrix
- Classification Report

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6. Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

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7. Model Evaluation

Metrics used:

- Accuracy
- Precision
- Recall

- F1-score

The model performs well due to structured dataset and clear logic.

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8. Visualization

A decision tree visualization is generated and saved as:

model_visualization.png

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9. Results

- Model predicts pass/fail accurately
- Important factors:
    - Study hours
    - Attendance
    - Assignments

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10. Advantages

- Easy to understand
- Fast execution
- Interpretable model

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11. Limitations

- Synthetic dataset
- Limited features
- Risk of overfitting

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12. Future Scope

- Use real-world data
- Apply advanced ML models
- Build web application
- Add more features

13. Conclusion

This project demonstrates how machine learning can improve educational outcomes by predicting student performance and helping institutions take early action.

14. References

- Scikit-learn Documentation
- Python Documentation
- ML Learning Resources